Scale Drawings

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lesson 1 — Scale Drawings	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.GR.1.5

2. Standards

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must. Find the scale factor between two figures (k = copy ÷ original, in either direction of size) and use a scale factor or a scale to find a length in either direction; recognise that angles are unchanged. (MA.7.GR.1.5, clarifications 1 and 3.)

Must not. Perimeter and area under scaling are 5.02–03 (k^2); floor plans are 5.04; dilations on the coordinate plane are MA.8.GR.2.2; similar triangles by name are Unit 6.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I: the same pair of figures as a picture, two side lengths and one number k
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	boards 1–3: one question three times — k over 1, under 1, a fraction

4. Learning Target and Essential Question

Learning target. I can find the scale factor between two figures, and use a scale to find a real length from a drawing or a drawing length from the real one.

Essential question. If a copy's sides are 6 and 9 where the original's were 4 and 6, what one number was every side multiplied by?

Building on. Grade 6 (MA.6.AR.3.2) — a ratio as a unit rate; Unit 4 — multiplying and dividing decimals and fractions.

Working toward. 5.02–03 — perimeter by k, area by k^2; 5.04 — floor plans and maps; Unit 6 — similar triangles are scale copies.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	13	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 5 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Two rectangles are 4 by 6 and 8 by 10. Why is the second not a scale copy of the first?	DOK 3
A copy is smaller than its original. What do you know about k before you compute it?	DOK 2
Every side of a triangle is tripled. Why can none of its angles change?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A 2 cm side becomes 6 cm in the copy. What is the scale factor?	***k = 3***	⅓ — divided original by copy [B1G-M 7.GR.1.5: the proportion set up backwards]; 4 — subtracted [7.GR.1.5: the additive error — Math Nation's named misconception]
2	An 8 cm side becomes 4 cm in the copy. What is the scale factor?	***k = ½***	2 — divided original by copy; 'k is always bigger than 1' [B1G-M 7.GR.1.5 named misconception]
3	A 12 cm side becomes 9 cm in the copy. What is the scale factor?	***k = ¾***	1⅓ — original ÷ copy [7.GR.1.5]; 3 — subtracted [7.GR.1.5: the additive error — Math Nation's named misconception]
4	A side of 5 cm becomes 2 cm in the copy. What is the scale factor?	***B — 0.4***	(A) original ÷ copy: 5 ÷ 2 [B1G-M 7.GR.1.5: the proportion set up backwards; 'k is always greater than 1']; (C) 'the same shape, so k = 1' [Math Nation 5.3: a scale factor of 1 is the same size, not the same shape]; (D) 5 − 2: the additive error — 'to get from 5 to 2, take away 3' [B1G-M 7.AR.3.3: 'just add' as a conversion]
5	The scale factor is 3.5. A side of the original is 6 cm. How long is that side in the copy?	***21 cm***	1.7 — divided by k going to the copy [Notes III: to the copy, multiply]
6	A drawing has a 40° angle. The real figure is a scale copy at $k = 2.5$. What is the angle in the real figure?	***A — 40°***	(B) 40 + 2.5: the additive error [7.GR.1.5: Math Nation's named misconception]; (C) 40 ÷ 2.5: the angle divided [7.GR.1.5]; (D) 40 × 2.5: the angle scaled like a length [7.GR.1.5: a scale factor multiplies lengths; the shape — its angles — is what stays the same]
7	A triangle has sides 6, 8 and 10. Another has sides 9, 12 and 15; a third has sides 9, 12 and 14. Which one is a scale copy of the first?	***The 9-12-15 triangle (k = 1.5); the 9-12-14 one is not a copy***	'both' — checked one pair and stopped [Notes I: the same k from every side]; 'neither' — compared 9 to 10 instead of corresponding sides
8	Dev, who is 13, draws a 15-ft wall as 3 in long. How long is a 20-ft wall on his drawing?	***4 in***	100 — multiplied by 5 going to the drawing [Notes III: back to the drawing, divide]; 8 — 20 − 15 + 3, the additive error
9	Priya says the scale factor from a 6 cm side to its 3 cm copy is 2. Is she right? Explain, and give the scale factor.	***No. k = 3 ÷ 6 = ½: the copy is smaller, so k is under 1.***	'Yes, because 6 is 2 times 3' — the proportion backwards [B1G-M 7.GR.1.5]; 'No, it is 3' — subtracted

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. The direction is written on the board all period as one line — copy ÷ original — and every k is found by writing the copy's number over the original's before dividing. The two rectangles of Notes I stay up beside it.

ELL. **Scale factor** is one number with no unit; **scale** has units on it (1 in : 20 mi). Say "copy divided by original" with every k, and point to which figure is which.

Enrichment. Board 7 finishes early for some: ask them for a fourth triangle with sides that are a scale copy of 6-8-10 at k = 2.5, and whether its angles changed.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say which way the division goes (copy ÷ original) and give k = ½. A board that says 'yes, 6 is 2 times 3' has the direction backwards — the unit's first misconception, and tomorrow's warm-up Q1 asks it again.

Vocabulary. **scale drawing**; **scale factor**; **corresponding sides**; **scale**

Materials. Whiteboards and markers. Two squares of different sizes cut from paper, for board 6's split: four right angles each. Calculators are allowed on everything and printed on nothing.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 5 test: Scale drawings: find the scale factor, missing side, or missing angle — WEA.